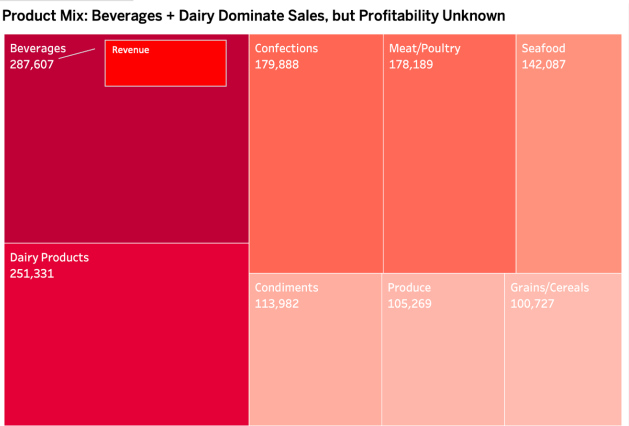
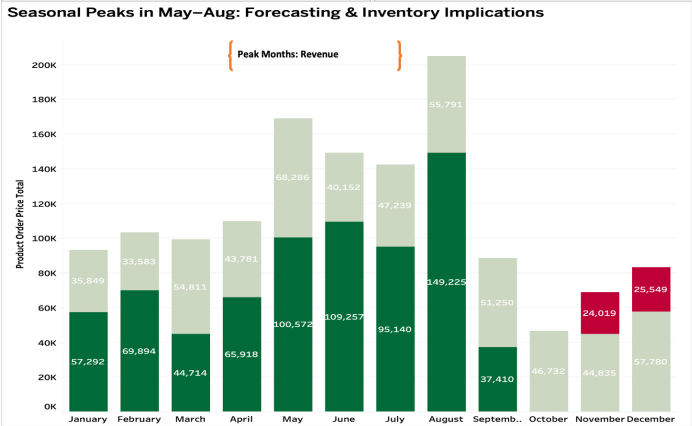
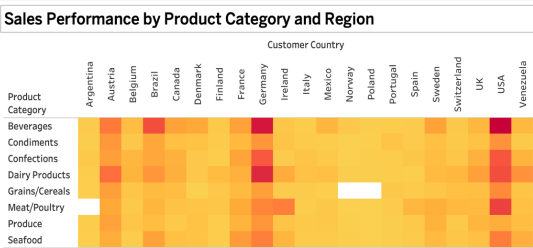
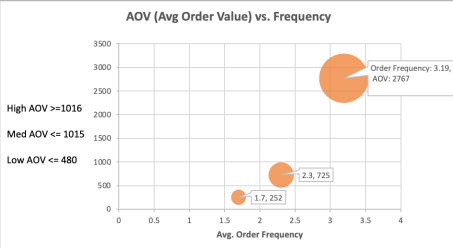
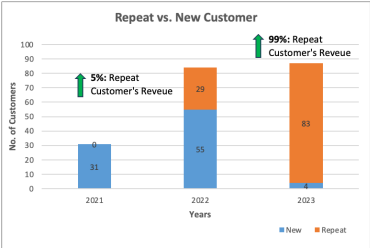


Executive Summary

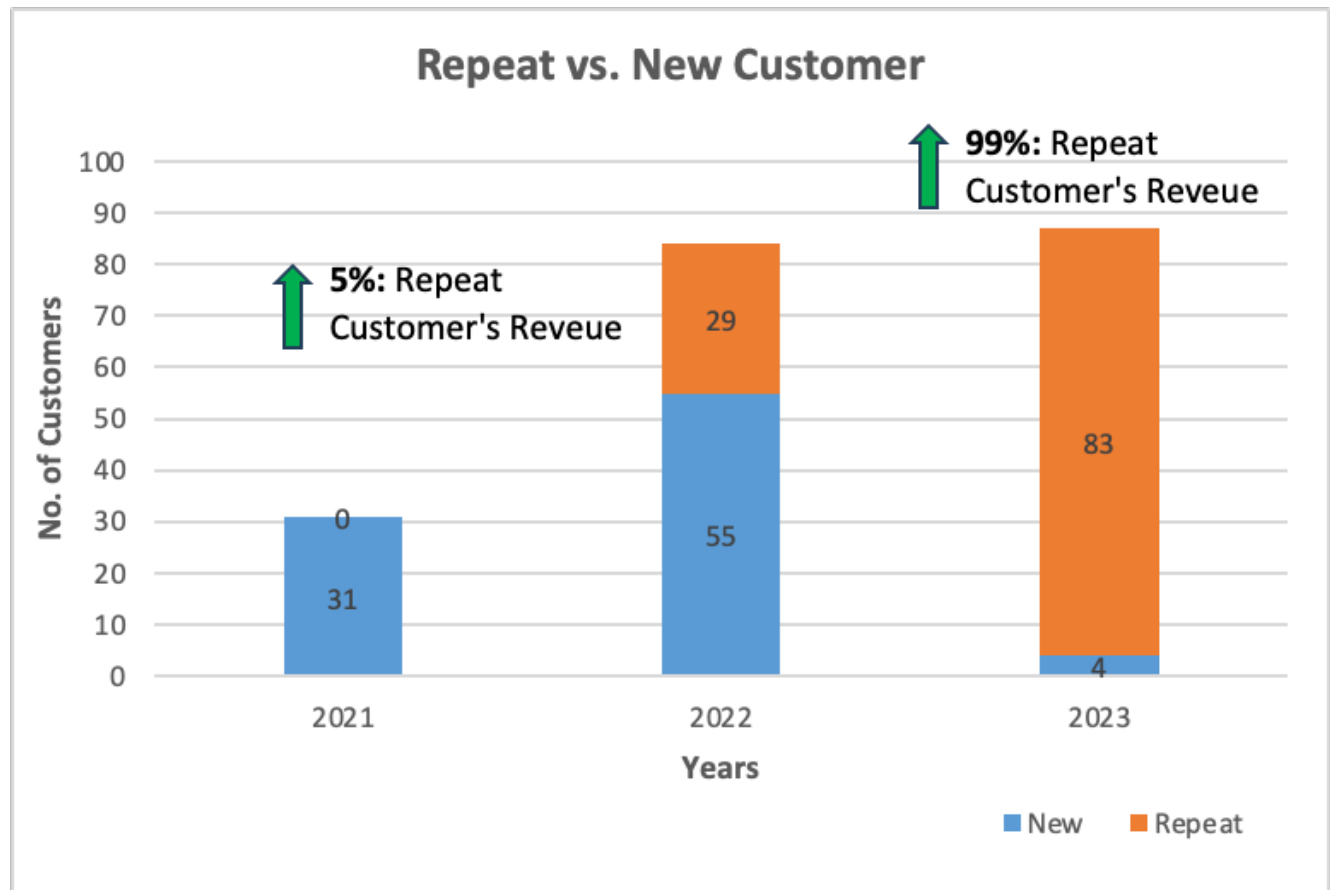
Northwind Traders' sales data from 2021 to 2023 highlights key business dynamics shaping performance and opportunity. Repeat customers now generate the majority of revenue, underscoring strong loyalty but signaling a decline in new customer acquisition. Average Order Value (AOV) and purchase frequency show segmentation potential, revealing high-value but under-engaged customers. Beverages and Dairy products dominate the product mix, though profitability by category remains unclear due to missing cost data. Seasonal sales peaks in May–August offer clear planning advantages, while sales by category and region suggest tailored strategies for different markets. Together, these insights support strategic decisions in customer retention, product focus, and regional prioritization.

Overview Visualization: High-Level Sales Dashboard



Deep Dive Analyses

1. Repeat vs. New Customer Analysis



Why this data subset?

To distinguish between customer acquisition and retention trends, which impact both marketing and revenue strategy.

What story does the visual tell?

Repeat customers made up a growing share of revenue year-over-year, while new customer acquisition declined steadily between 2021 and 2023.

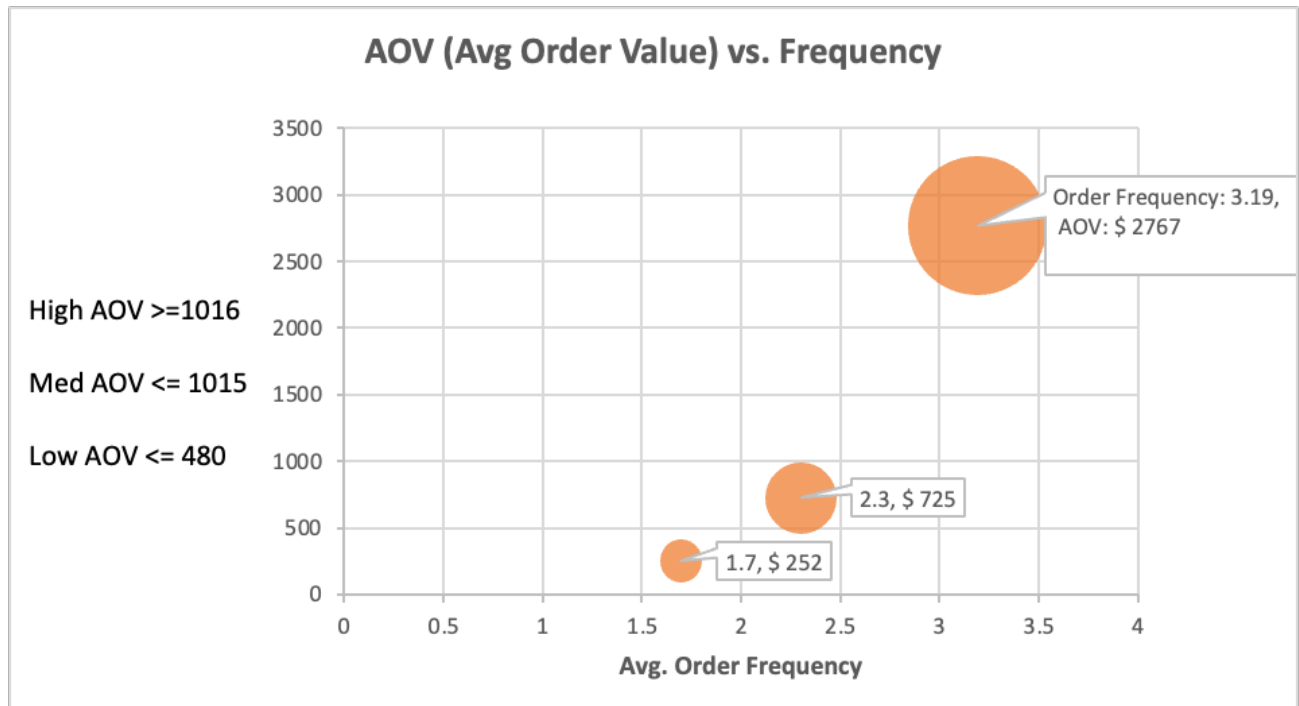
What do we now know?

Customer loyalty is strong, but acquisition efforts are weakening, which could hinder future growth.

What don't we know but need to know?

The reasons behind the drop in new customer acquisition are market saturation, reduced marketing spend, or operational barriers

2. Average Order Value (AOV) vs. Frequency



Why this data subset?

Identifies behavioral customer segments based on spending and engagement.

Supports targeted marketing, bundling, and operational resource allocation decisions.

What story does the visual tell?

High-AOV customers spend the most (\$2,767) but purchase only ~3 times in 3 years; it could be more.

High-AOV customers are 50% of the overall customer base.

Low-AOV customers order fewer and spend less (1.7, \$252).

Medium-AOV customers show a balanced pattern of spend and frequency.

What do we now know?

High-AOV customers are prime candidates for engagement campaigns to boost order volume.

What don't we know but need to know?

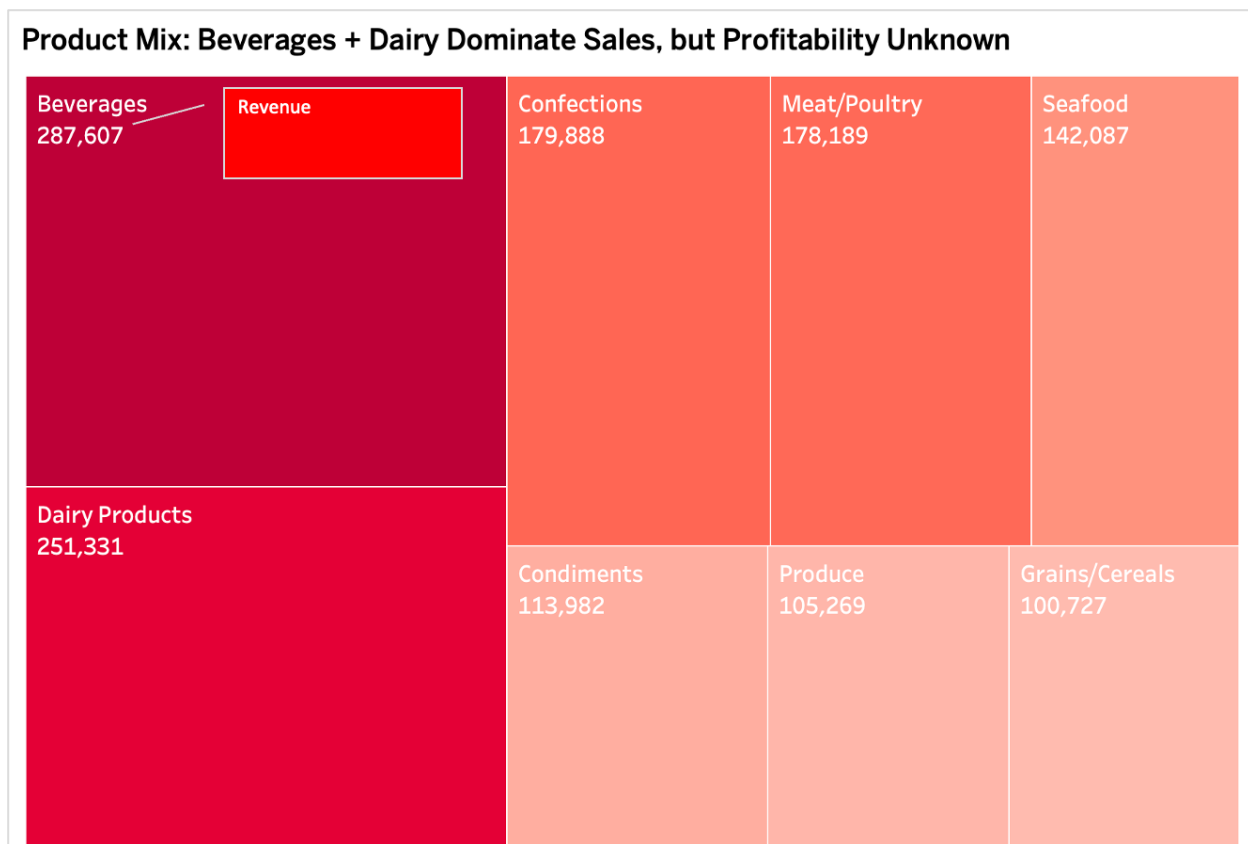
What motivates each segment (e.g., product type, discounts, urgency)?

Are high-AOV customers loyal or price-sensitive occasional buyers?

What incentives or campaigns could increase frequency among high spenders?

How do customer satisfaction, acquisition cost, and lifetime value vary across segments?

3. Product Mix – Beverages + Dairy Dominate



Why this data subset?

To understand what product categories contribute most to revenue and identify opportunities for optimization.

What story does the visual tell?

Beverages and Dairy consistently lead in total revenue, suggesting strong demand and volume sales across regions.

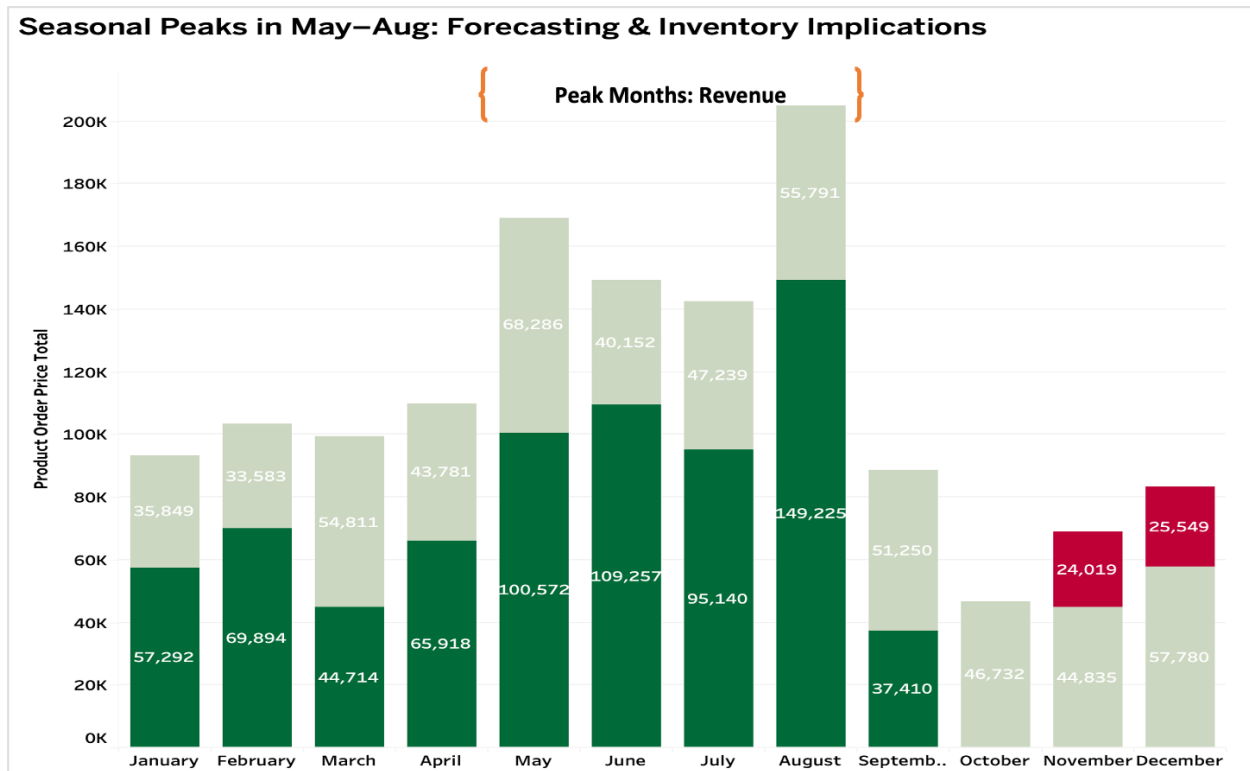
What do we now know?

Sales are concentrated in a few core categories, but without cost data, we can't assess their true contribution to profitability.

What don't we know but need to know?

Per-category profit margins, product-level return rates, or waste/spoilage rates that could drastically alter perceived value.

4. Seasonal Peak in May–August



Why this data subset?

To support forecasting, inventory, and marketing planning by identifying high and low sales periods.

What story does the visual tell?

Sales consistently spike in the summer months, particularly May to August, and drop in October–November.

What do we now know?

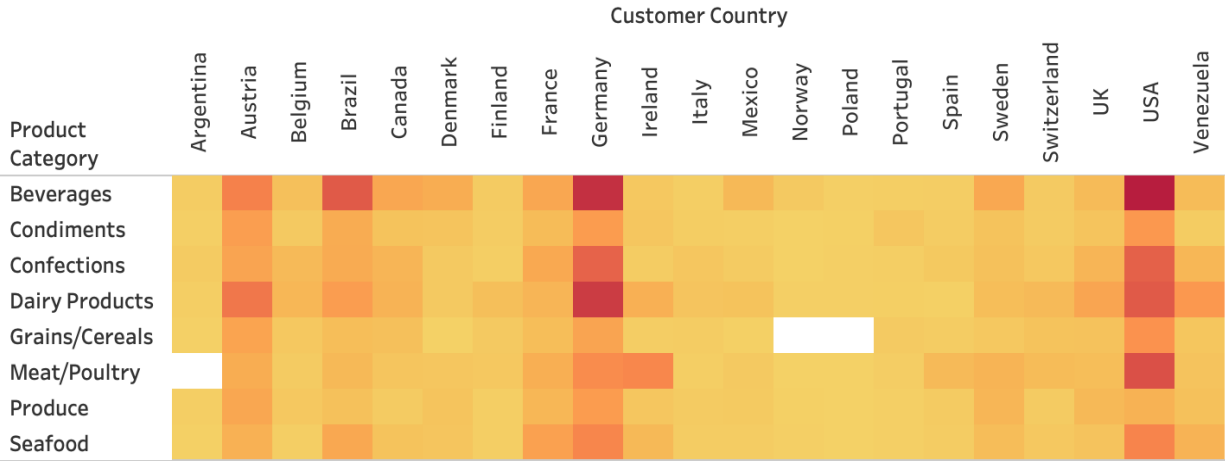
There is predictable seasonality that should be leveraged for inventory buildup, promotional campaigns, and resource planning.

What don't we know but need to know?

How seasonality differs across product categories or regions, and whether this trend is driven by internal campaigns or external demand cycles.

5. Sales Performance by Category and Region

Sales Performance by Product Category and Region



Regional Sales Distribution: 32% Orders & 37% Revenue Comes from Two Countries (USA & Germany)



Why this data subset?

To discover how product performance varies by geography and identify region-specific strengths and gaps.

What story does the visual tell?

Certain categories dominate in some regions (e.g., Beverages in the U.S., Confections in Europe), revealing clear regional preferences.

What do we now know?

Product success is not uniform — regional preferences should guide inventory, pricing, and promotional strategies.

What don't we know but need to know?

Whether these differences are due to consumer demand, supply chain efficiency, or sales strategy, and how to optimize region-category alignment.

Conclusions

What is known

- Revenue growth is now being driven more by **repeat customers** than new ones.
- **High-AOV customers** are not being fully leveraged, while operational load may be heavy on low & mid-AOV buyers.
- **Beverages and Dairy** lead in volume, but profitability by category is unclear.
- **Seasonal sales trends** are consistent and actionable.
- Product performance **varies by region**, suggesting potential for localization.

What is unknown

- The **profitability** of top categories (cost/margin data missing)
- Why **new customer acquisition** is falling
- What drives **regional product preferences**
- Whether **seasonal trends** apply equally across product and market segments

Next Steps

1. Integrate **Activity-Based Costing (ABC)** to uncover product and customer-level profitability.
2. Investigate **churn reasons and marketing funnel performance** to revive acquisition.
3. Conduct **regional analysis of inventory, promotions, and fulfillment** strategies.
4. Expand the analysis to include **category-level seasonal trends** and customer engagement metrics.

Recommendations to SLT

- **Prioritize retention** and enhance orders of high-AOV customers with loyalty and bundling strategies.
- **Reinvest in acquisition** via digital and channel marketing to reverse declining new customer trends. 99% of 2023 revenue came from returning customers - a sign of strong loyalty. However, new customer acquisition dropped 93%, suggesting a weakening marketing funnel that needs high attention
- Launch a **pilot localization initiative** — adjusting product focus and marketing based on regional sales behavior.
- Implement **ABC costing** to better align decisions with profit, not just volume.